

Automating EC2 with AWS Lambda Using Boto3 – Lab 2

@Amel

Definition

- Amazon explains AWS Lambda (λ) as a 'serverless' compute service, meaning the developers, don't have to worry about which AWS resources to launch, or how will they manage them, they just put the code on lambda and it runs, it's that simple! It helps you to focus on core-competency i.e. App Building or the code.
- AWS Lambda executes your backend code, by automatically managing the AWS resources. When we say 'manage', it includes launching or terminating instances, health checkups, auto-scaling, updating or patching new updates etc.

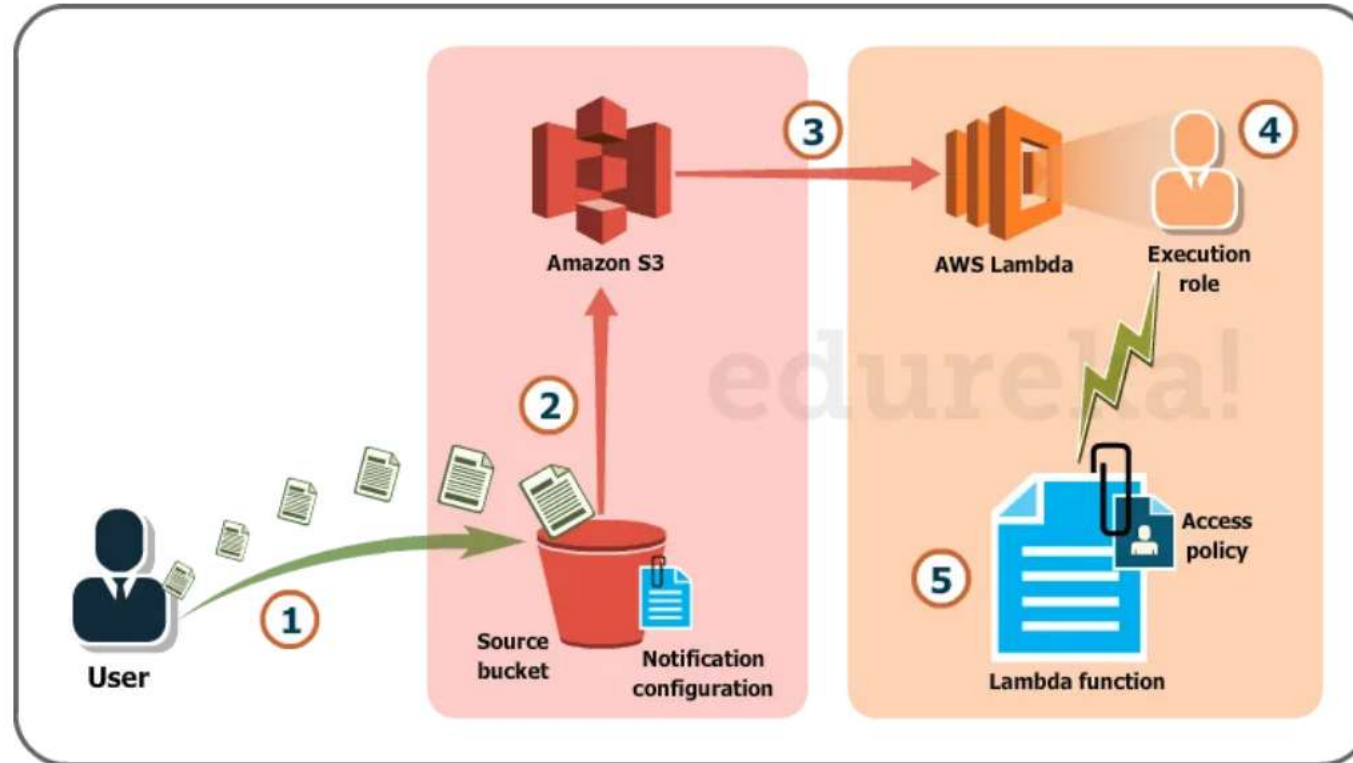
How it works?

- The code that you want Lambda to run is known as a **Lambda function**. Now, as we know a function runs only when it is called, right? Here, **Event Source** is the entity which triggers a Lambda Function, and then the task is executed.
- Different types of Event Source are available, based on that Lambda functions are called and perform the activity as per the flow
- Serverless architecture is used in other way in Kubernetes using knative api

Example

- Suppose you have an app for image uploading. Now when you upload an image, there are a lot of tasks involved before storing it, such as resizing, applying filters, compression etc.
- So, this task of uploading of an image can be defined as an **Event Source** or the 'trigger' that will call the Lambda Function, and then all these tasks can be executed via the Lambda function.
- In this example, a developer just has to define the event source and upload the code.

• •



explanation

1. User uploads an image (object) to a source bucket in S3 which has notification attached to it, for Lambda.
2. The notification is read by S3 and it decides where to send that notification.
3. S3 sends the notification to Lambda, this notification acts as an invoke call of the lambda function.
4. Execution role in Lambda can be defined by using IAM (Identity and Access Management) to give access permission for the AWS resources, for this example here it would be S3.
5. Finally, it invokes the desired lambda function which works on the object which has been uploaded to the S3 bucket.

Challenges with Traditional approach

- *If you were to solve this scenario traditionally, along with development, you would have hired people for managing the following tasks:*
- Size, provision and scale up group of servers
- Managing OS updates
- Apply security patches and
- Monitor all this infrastructure for performance and availability.

Pricing in AWS Lambda

- Like most of the AWS services, AWS Lambda is also a pay per use service, meaning you only pay what you use, therefore you are charged on the following parameters
- The number of requests that you make to your lambda function
- The duration for which your code executes.
- AWS Lambda counts a request each time it starts executing in response to an event source or invoke call, including test is invoked from the console. Let's look at the prices now:
- First 1 million requests, every month are for free.
- 0.20\$ per million requests thereafter.

Aws console

Amazon Web Services

Compute



EC2

Virtual Servers in the Cloud



EC2 Container Service

Run and Manage Docker Containers



Elastic Beanstalk

Run and Manage Web Apps



Lambda

Run Code without Thinking about Servers



Server Migration

Migrate on-premises servers to AWS

Storage & Content

Developer Tools



CodeCommit

Store Code in Private Git Repositories



CodeDeploy

Automate Code Deployments



CodePipeline

Release Software using Continuous Delivery

Management Tools



CloudWatch

Monitor Resources and Applications



CloudFormation

Create and Manage Resources with Templates

Internet of Things



AWS IoT

Connect Devices to the Cloud

Game Development



GameLift

Deploy and Scale Session-based Multiplayer Games

Mobile Services



Mobile Hub

Build, Test, and Monitor Mobile Apps

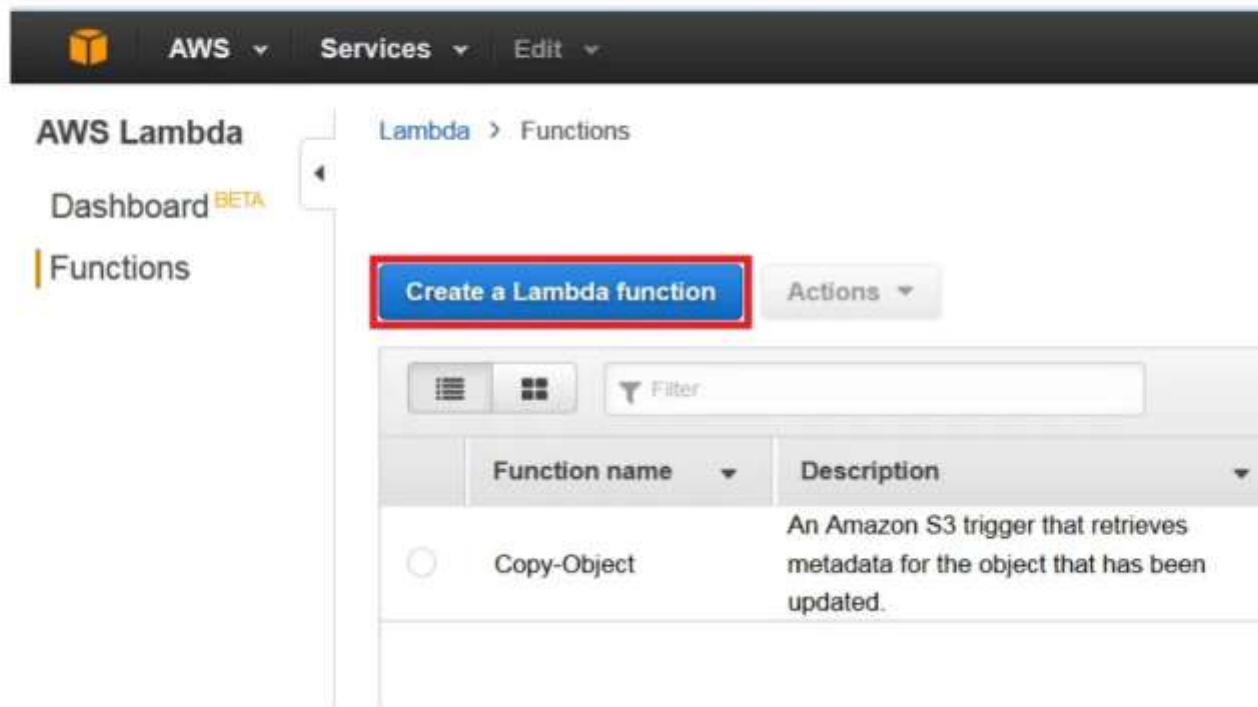


Cognito

User Identity and App Data Synchronization

• •

Step2: On the AWS Lambda Console, click on “Create a Lambda function”.



• •

 **AWS** ▾ **Services** ▾ **Edit** ▾

[Lambda](#) > [New function](#)

Select blueprint

[Configure triggers](#)

[Configure function](#)

[Review](#)

Select blueprint

Blank Function

Configure your function from scratch.
Define the trigger and deploy your code
by stepping through our wizard.

custom

s3-get-object-python

An Amazon S3 trigger that retrieves
metadata for the object that has been
updated.

python2.7 · s3

Configure Trigger

Configure triggers

Configure an optional trigger to automatically invoke your function.

1.



 Lambda

Remove

2.

Filter integrations

 AWS IoT

 S3

 SNS

Cancel

Previous

3. Next

Feedback

English

ates. All rights reserved. Privacy Policy Terms of Use

• •

Configure function

A Lambda function consists of the custom code you want to execute. [Learn more](#) about Lambda functions.

Name*

ex-lambda

Description

A function to log the addition of an object in

Runtime*

Node.js 4.3

Lambda function code

Code entry type

Edit code inline

```
1 exports.handler = (event, context, callback) => {  
2   console.log('An object in S3 is added');  
3 };
```

• •

Lambda function handler and role

Handler*

Role* ⓘ

Role name* ⓘ

Policy templates ⓘ

* These fields are required.

[Cancel](#)

[Previous](#)

[Next](#)

[Cancel](#)

[Previous](#)

[Create function](#)

• •

Step7: Now, since we created the function for S3 bucket, the moment you add a file to your S3 bucket, you should get a log for the same in CloudWatch, which is a monitoring service from AWS.

Filter events		all 30s 5m 1h 6h 1d 1w custom ▾
Time (UTC +00:00)	Message	
2016-10-25		
No older events found at the moment. Retry .		
▶ 15:18:08	START RequestId: 3bdd7540-9ac6-11e6-ba06-f9f5aabe9def Version: \$LATEST	
▼ 15:18:08	2016-10-25T15:18:08.416Z 3bdd7540-9ac6-11e6-ba06-f9f5aabe9def An object in S3 is added	
2016-10-25T15:18:08.416Z 3bdd7540-9ac6-11e6-ba06-f9f5aabe9def An object in S3 is added		
▶ 15:18:08	END RequestId: 3bdd7540-9ac6-11e6-ba06-f9f5aabe9def	
▶ 15:18:08	REPORT RequestId: 3bdd7540-9ac6-11e6-ba06-f9f5aabe9def Duration: 17.23 ms Billed Duration: 100 ms Memory Size: 128 M	
No newer events found at the moment. Retry .		

• •

- Lambda with s3
- Lambda with api-gateway